How Much PENCIL Can You Afford? Compute-allocation for link prediction

DA4022 mid-project update | Amit Kumar · Ashmit Satpute · Arnav Kshirsagar · Pranav Shalya · Atharv Wankhede

WHAT WE BUILT

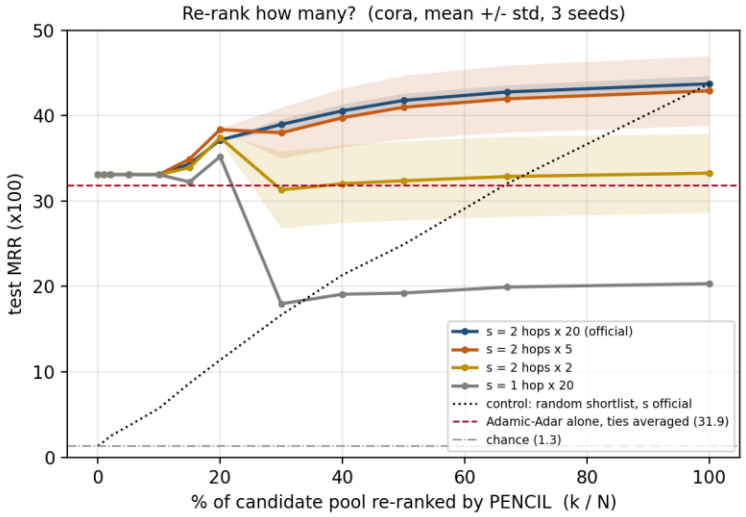
- Reproduced PENCIL on one free T4: test MRR 44.0 (paper: 42.2).
- Retrieve-then-rerank: Adamic–Adar (AA) ranks every candidate; PENCIL re-scores only the top k.
- Two inference knobs, no retraining: k = how many re-ranked; s = sampling budget per candidate (hops × neighbours).
- Cora: 6 settings of s × 3 seeds × 12 values of k. Cost = measured T4 forward time.
- Checks: our AA/CN reproduce the HeaRT benchmark exactly (31.87 / 20.99); our metric matches the repo evaluator ($\Delta < 10^{-7}$).

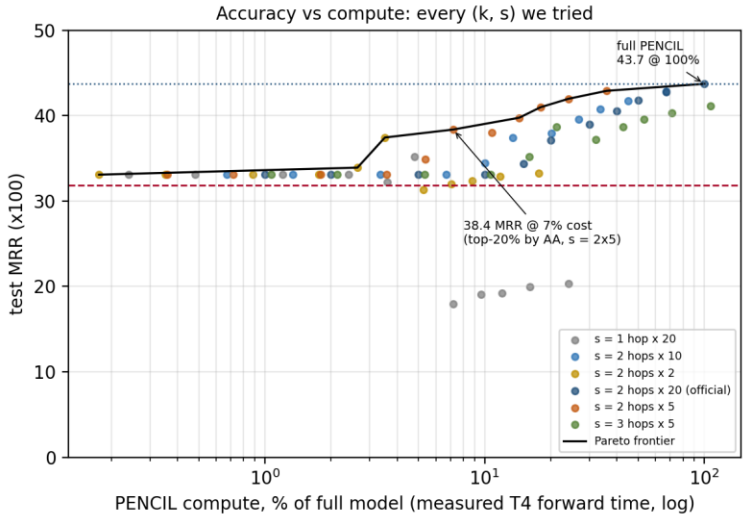
FINDINGS SO FAR

- **Cheap regime:** re-ranking AA's top 20% at s = 2×5 costs 7% of full PENCIL and adds +5 MRR over AA on both splits.
- **Halving s is free:** s = 2×10 matches full PENCIL on test and validation at 67% of the compute. s = 2×5 is cheaper but unstable (seed std up to 4.1).
- **Re-ranking the top 10% does nothing:** AA's top picks are already true links. PENCIL earns its cost only where AA is blind: 57% of true test links share no neighbour.
- **k and s interact:** a weak re-ranker (1 hop) peaks at k = 20% and then hurts (35 → 20 MRR); the full one keeps improving.

NEXT (→ Nov 26)

- citeseer + ogbl-collab, and HeaRT per-query hard negatives.
- Fit a (k, s) rule on validation; test it on a graph it never saw.
- Proper latency and memory benchmark (repeats, batch size 1).
- Training axis: 93% of final MRR is reached by epoch 42 of 150.





Setting (cora)	test MRR	valid MRR	compute
Adamic–Adar alone (degree tie-break)	33.1	38.1	0%
AA top-20% → PENCIL, s = 2×5	**38.4 ± 0.1**	**43.1 ± 0.1**	**7%**
All → PENCIL, s = 2×10	**42.9 ± 1.9**	**59.7 ± 0.5**	**67%**
All → PENCIL, s = 2×5	42.9 ± 4.1	52.4 ± 2.8	36%
Full PENCIL (official s = 2×20)	43.7 ± 0.9	59.8 ± 0.3	100%

Protocol: cora planetoid split; each of 527 test links ranked against all 527 test negatives (chance MRR = 1.3); validation 263 vs 263. Compute = measured T4 forward time vs full PENCIL. Settings above are read off a sweep, not yet selected on validation alone — the final report will. One graph so far; timings from a single pass.